Audio Control Panel – General Data

1. General

The audio control panels (KMA 29) is used to make audio selections for all aircraft audio communications. It includes an automatic voice activated (VOX) intercommunication design, which eliminates the requirement for intercom squelch adjustment.

2. Description

A. Audio Control Panel (KMA 29)

Refer to Figure 1 for details.

The audio control panel supports up to five communication radios, one of which can be configured as a telephone (full duplex, with no requirement to push to talk). It also supports up to eight switched audio inputs (VHF navigation, COM, ADF, DME, marker beacon, and auxiliary). The panel also provides an interface to the Passenger Address (PA) system amplifier and the aural warning system. A marker beacon receiver is integral to the panel.

An audio control panel is installed above each PFD. Momentary push-buttons are used to select the desired COM transceivers. Pressing a button turns on the associated receiver and the green LED. The pilot can identify which receivers are selected by noting which LEDs are on. The left and right audio panels and left and right crew headsets operate independently of each other. Each pilot may transmit and receive at the same time, using different VHF or other communication radios such as HF or satellite telephone.

A marker beacon receiver provides the necessary marker beacon signals to the PFD and audio indications for an instrument Landing system (ILS). The MKR push button allows the MKR SENS button to be used to set either high or low sensitivity or mute the marker beacon audio by pressing the MUTE button on the pilots panel for five seconds. The marker beacon system may only be tested from the left audio panel. The test function on the right side (co-pilot) audio panel is inoperative.

Each audio panel has a single-button emergency reversion function that allows the pilot headset to be directly connected to the number 1 communications radio. The audio panel also supports digital data communication between the two panels, and between the audio panels and the CVR.

The KMA 29 has an automatic dimming of the pushbutton annunciator LEDs controlled by a photocell. Control of the unit backlighting is through the ‘INSTRUMENT PANEL' dimmer on the ‘DIMMING PANEL' installed on the center control column.

28 Volt DC power is protected by a 5 amp PILOT AUDIO (3E) (Left Bus) and a 5 amp COPILOT AUDIO (4E) (Right Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The KMA 29 audio control panel interfaces to the following aircraft systems:

1) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

2) VHF Communication System (refer to 23-12-00, VHF Communications System – General Data).

3) VHF Navigation System (refer to 34-52-00, VHF Navigation System – General Data).

4) HF Communication System (refer to 23-11-00, HF Communication System (S.O.O. 6215) – General Data).

5) Distance Measuring Equipment (DME) (refer to 34-54-00, Distance Measuring Equipment (DME) (Mod 6/2049) – General Data).

6) Cabin Passenger Address (PA) (refer to 44-00-00, Cabin Systems – General Data).

7) Cockpit Voice Recorder (CVR) (refer to 31-32-00, Cockpit Voice Recorder (CVR) – General Data).

8) Audio System Isolation Amplifier (refer to 23-54-00, Audio System Isolation Amplifier – General Data).

9) Noise Cancelling Headsets (refer to 23-52-00, Noise Cancelling Headsets – General Data).

10) VHF-FM (refer to 23-13-00, VHF-FM System (S.O.O. 6296) – General data).

3. Adjustment/Test

A. Function Test Audio Control Panel (KMA 29)

1) Intercommunication System (ICS) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that PILOT AUDIO (3E) and COPILOT AUDIO (4E) circuit breakers are closed on the avionics circuit breaker panel.

d) Connect a headset to the pilot, co-pilot and observers.

e) Turn the intercom system (ICS) control to the middle position.

f) Speak into the pilot, co-pilot and observer positions. Ensure intercommunication at each position is satisfactory.

g) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Receive/Transmit Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure the PILOT AUDIO (3E) and COPILOT AUDIO (4E) circuit breakers are closed on the avionics circuit breaker panel.

d) Connect a headset to the pilot, and/or co-pilot position.

e) Set MIC selector to an unused COM position. Ensure all audio selector LEDs are OFF, except the selected transmit position.

f) Tune radio to receive a valid audio signal. Select associated receive audio on the audio control panel.

g) Verify the selected pilot audio control panel receivers are present on the pilot's headset and that the selected co-pilot control panel audio are present on the co-pilot's headset.

1 COM 1

2 COM 2

3 COM 3 – if used

4 COM 4 – if used

5 COM 5 – if used

6 NAV 1

7 NAV 2

8 ADF 1 – if installed

9 ADF 2 – if installed

10 DME 1

11 DME 2 – if installed

h) Tune radio to an appropriate transmit test ground, tower, or base station.

i) Select the associated MIC on the pilot's audio control panel and/or the selected co-pilot's control panel. Do a transmit radio check for each radio.

1 COM 1

2 COM 2

3 COM 3 – if used

4 COM 4 – if used

5 COM 5 – if used

j) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Operational Test Audio Panel Emergency Switch

1) In the OFF or EMG position, the pilot is connected directly to COM 1. This allows communication capability regardless of unit condition. Any time power is removed or turned OFF, the audio selector will be placed in the fail-safe mode. In EMG mode, critical audio signals on Unswitched 1 and Unswitched 2 are also provided to the pilot headset. In EMG mode, audio signals are present in only one ear of a stereo headset.

2) Complete a radio TX/RX test. Ensure proper transmit and receive functions of COM 1.

4. Removal/Installation

These procedures are applicable to the removal and installation of the pilot's Audio Control Panel and the co-pilot's Audio Control Panel (KMA 29).

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND LIGHTNING. ENSURE INTEGRITY OF THE HIRF/LIGHTNING PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – PANEL/PEDESTAL MOUNT UNITS CANNOT BE RELIABLY GROUNDED TO THE AIRFRAME VIA THE PANEL MOUNTING SCREWS OR DZUS FASTENERS. GROUNDING OF THESE UNITS IS ACCOMPLISHED BY USING THE SHIELD GROUNDING BRAID OR JUMPER. VERIFY PROPER ELECTRICAL BONDING AS DETAILED IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELLS/STRAIN RELIEF AND THE CONNECTOR AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of the Audio Control Panel (KMA 29)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the PILOT AUDIO (3E) and/or the COPILOT AUDIO (4E) circuit breaker on the avionics circuit breaker panel and fit safety clips.

CAUTION **CARE MUST BE TAKEN TO ENSURE THE WRENCH IS SEATED CORRECTLY ON THE SCREW BEFORE ROTATING, UNIT CAN BE DAMAGED.**

c) Using a 3/32 hexagonal-head wrench, engage it into the audio control panel locking screw.

d) Turn the locking screw counterclockwise until it disengages from the audio control panel mount.

e) Pull on the edges of the audio panel, until the audio control panel is removed from the mount.

f) Remove unit from aircraft.

Note Stow connectors as required.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the PILOT AUDIO (3E) and/or the COPILOT AUDIO (4E) circuit breaker on the avionics circuit breaker panel is opened and fitted with safety clips.

c) Slide the audio panel controller into the mount on the instrument panel.

d) Push carefully on the face of the audio control panel until the locking screw is engaged at the rear of the mount.

CAUTION **CARE MUST BE TAKEN TO ENSURE THE WRENCH IS SEATED CORRECTLY ON THE SCREW BEFORE ROTATING, UNIT CAN BE DAMAGED.**

e) Using a 3/32 hexagonal-head wrench, engage it into the audio control panel locking screw.

f) Apply a light pressure to the edges of the unit, while turning the 3/32 wrench clockwise to tighten the locking screw.

g) Remove safety clips and close PILOT AUDIO (3E) and/or the COPILOT AUDIO (4E) circuit breaker on the avionics circuit breaker panel.

h) Test the audio control panel as per Para 3.A.

5. Troubleshooting

A. Audio Control Panel (KMA 29)

1) If the Audio Control Panel (KMA 29) is deemed inoperative check the following:

a) Ensure PILOT AUDIO (3E) and the COPILOT AUDIO (4E) circuit breakers are closed.

b) Check that the units are securely installed and the locking screw is securely tightened.

c) Visually inspect audio control panel connectors for damaged wires and contacts.

d) Ensure correct position of the PTT switch. Visually inspect PPT switch for damaged wires or contacts.

e) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

f) If repair or service is required, send to appropriate rated avionics repair facility.

2) Factory Configuration

a) Indication arrows above the microphone selectors indicate which side has which radio selected for transmit. Annunciation of which radio has been selected for transmission at the off-side position can be configured to enabled or disabled. When the offside indication is disabled, the arrow for the Mic selected will not be lit for the opposite KMA 29 position. When enabled, the pilot can see which radio the co-pilot has selected for transmit, and vice versa, by noting which of the arrows is illuminated. Each KMA 29 radio can be configured separately.

The factory configuration is that each audio panel have off-side transmit selection indication enabled. Set to approved factory default, as described below, if this feature is found to be disabled.

To toggle the offside transmit selection indication:

1 Press the right side of the “SPLIT” button three (3) times within one and a half seconds (1 ½) to toggle.

Note If the mode has been enabled, the NAV 1 indicator will blink once. If the mode has been disabled, the NAV 1 indicator will blink twice.

b) The KMA 29 contains five separate backlighting curves. The factory configuration for the backlighting curve is curve 3, which is achieved by following the procedure described below.

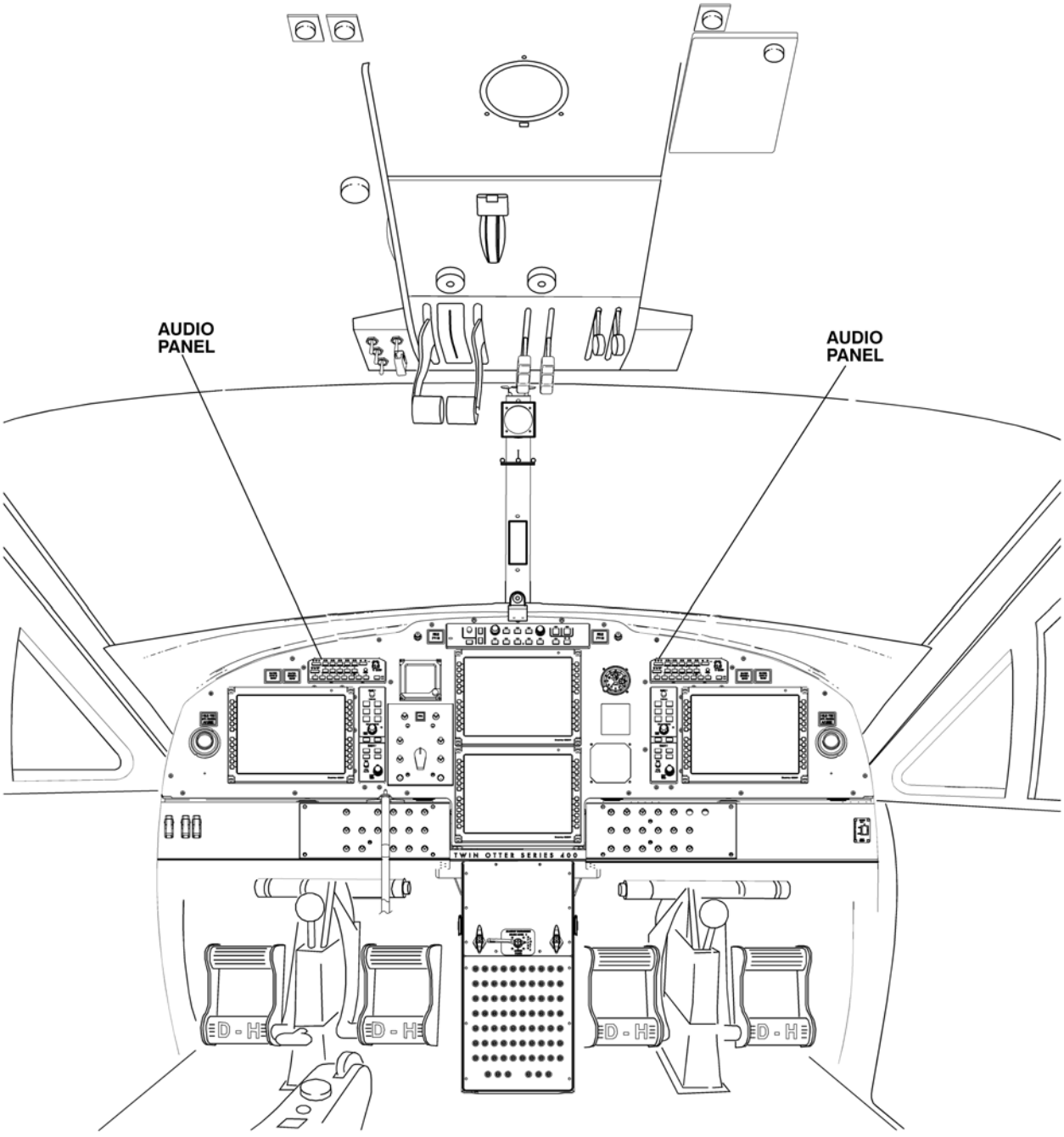
To select the factory approved backlighting curve:

1 Press and hold left split for 10 seconds.

Note Left split LED will start to blink.

2 Press COM 3 XMIT to select the approved factory default lighting curve.

Note The unit will exit this mode either after a COM is pressed or 10 seconds.



Audio Control Panels
Figure 1